1. $\lim\limits_{x\to 1}\dfrac{\sqrt[n]{x}-1}{\sqrt[k]{x}-1}$; $n,k\in\mathbb{N}$

$$\lim_{x\to 1}\frac{\sqrt[n]{x}-1}{\sqrt[k]{x}-1}=\lim_{x\to 1}\frac{(1+(x-1))^{\frac{1}{n}}-1}{(1+(x-1))^{\frac{1}{k}}-1}=\lim_{x\to 1}\frac{\frac{1}{n}(x-1)}{\frac{1}{k}(x-1)}=\frac{k}{n}$$

2. $\lim\limits_{x\to\pi}\dfrac{1-\cos x\cos(2x)\cos(3x)}{1+\cos x}=\Big/\begin{matrix}t=x-\pi\\ t\to 0\end{matrix}\Big/=\lim\limits_{t\to 0}\dfrac{1-\cos t\cos 2t\cos 3t}{1-\cos t}=$

$$=\lim_{t\to 0}\frac{2-\cos 3t(\cos t+\cos 3t)}{t^2}=\lim_{t\to 0}\frac{1+\sin^2 3t-\cos t\cos 3t}{t^2}=$$

$$=\lim_{t\to 0}\frac{\sin^2 3t}{t^2}+\lim_{t\to 0}\frac{1-\cos t\cos 3t}{t^2}=9+\lim_{t\to 0}\frac{1-\cos^2 t(1-4\sin^2 t)}{t^2}\ (=)$$

$\triangle\ \cos 3t=\cos 2t\cos t-\sin 2t\sin t=\cos t(1-2\sin^2 t-2\sin^2 t)$

$$(=)\ 9+\lim_{t\to 0}\frac{\sin^2 t+\sin^2 2t}{t^2}=9+\lim_{t\to 0}\frac{\sin^2 t}{t^2}+\lim_{t\to 0}\frac{\sin^2 2t}{t^2}=10+4=14$$

3. $\lim\limits_{x\to 0}\dfrac{1-\cos x\sqrt{\cos 2x}}{\operatorname{tg} x^2}=\lim\limits_{x\to 0}\dfrac{\sin^2 x+\cos^2 x-\cos x\sqrt{\cos 2x}}{x^2}=\Big/\sim -\dfrac{x^2}{2}\Big/$

$$=\lim_{x\to 0}\frac{\sin^2 x}{x^2}+\lim_{x\to 0}\frac{\cos x(\cos x-\sqrt{\cos 2x})}{x^2}=1+\lim_{x\to 0}\frac{\cos x-1}{x^2}-\lim_{x\to 0}\frac{-1+\sqrt{\cos 2x}}{x^2}=$$

$$= 1 - \frac{1}{2} - \lim_{x\to 0} \frac{(1+(-2\sin^2 x))^{\frac{1}{2}} - 1}{x^2} = \frac{1}{2} + \lim_{x\to 0} \frac{\sin^2 x}{x^2} = \frac{3}{2}$$

4. $$\lim_{x\to \frac{\pi}{2}} \frac{\sqrt[4]{\sin x} - \sqrt[3]{\sin x}}{\cos^2 x} = \left| \begin{matrix} t = x - \frac{\pi}{2} \\ t \to 0 \end{matrix} \right| = \lim_{t\to 0} \frac{\sqrt[4]{\cos t} - \sqrt[3]{\cos t}}{\sin^2 t} =$$

$$= \lim_{t\to 0} \frac{(1+(\cos t - 1))^{\frac{1}{4}} - 1 - ((1+(\cos t - 1))^{\frac{1}{3}} - 1)}{t^2} = \lim_{t\to 0} \frac{(1+\cos t - 1)^{\frac{1}{4}} - 1}{t^2} -$$

$$- \lim_{t\to 0} \frac{(1+(\cos t - 1))^{\frac{1}{3}} - 1}{t^2} = \frac{1}{4} \lim_{t\to 0} \frac{\cos t - 1}{t^2} - \frac{1}{3} \lim_{t\to 0} \frac{\cos t - 1}{t^2} = \frac{1}{12} \lim_{t\to 0} \frac{-\cos t + 1}{t^2} = \frac{1}{24}$$

5. $$\lim_{x\to\infty} \left(\sin\sqrt{x^2+1} - \sin\sqrt{x^2-1} \right) = \lim_{x\to\infty} 2\sin\left(\frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{2} \right) \cos\left(\frac{\sqrt{x^2+1} + \sqrt{x^2-1}}{2} \right) =$$

$$= \lim_{x\to\infty} 2\sin\left(\frac{1}{\sqrt{x^2+1} + \sqrt{x^2-1}} \right) \cos\left(\frac{\sqrt{x^2+1} + \sqrt{x^2-1}}{2} \right) =$$

$$= \lim_{x\to\infty} \frac{2}{\sqrt{x^2+1} + \sqrt{x^2-1}} \cos\left(\frac{\sqrt{x^2+1} + \sqrt{x^2-1}}{2} \right) \Leftarrow \lim_{x\to\infty} \frac{2}{\sqrt{x^2+1} + \sqrt{x^2-1}} = 0$$

$$\Leftarrow \lim_{x\to\infty} 0 \cdot \cos\left(\frac{\sqrt{x^2+1} + \sqrt{x^2-1}}{2} \right) = 0$$

6. $\lim\limits_{x\to 0} \dfrac{\pi - 4\operatorname{arcctg}\frac{1}{1+x}}{x} = \lim\limits_{x\to 0} \dfrac{4\left(\operatorname{arcctg} 1 - \operatorname{arcctg}\frac{1}{1+x}\right)}{x}$ (=)

$\operatorname{arcctg} x - \operatorname{arcctg} y = \operatorname{arcctg}\dfrac{x-y}{1+xy}$

(=) $\lim\limits_{x\to 0} \dfrac{4}{x}\cdot \operatorname{arcctg}\left(\dfrac{1-\frac{1}{1+x}}{1+\frac{1}{1+x}}\right) = \lim\limits_{x\to 0} \dfrac{4}{x}\operatorname{arcctg}\left(\dfrac{x}{x+2}\right) =$

$= \lim\limits_{x\to 0} \dfrac{4}{x}\cdot \operatorname{arcctg}\left(1 - \frac{2}{x+2}\right) = \lim\limits_{x\to 0} \dfrac{4}{x}\cdot\left(1-\dfrac{2}{x+2}\right) = \lim\limits_{x\to 0} \dfrac{4}{x}\cdot\dfrac{x}{x+2} =$

$= 2$

7. $\lim\limits_{x\to 1/4} \dfrac{1-\operatorname{ctg}(\pi x)}{\ln(\operatorname{tg}(\pi x))}$ (=); $\ln(\operatorname{tg}(\pi x)) = \ln\left(1+(\operatorname{tg}(\pi x)-1)\right) \sim \operatorname{tg}(\pi x)-1$ при $x\to\frac{1}{4}$

(=) $\lim\limits_{x\to 1/4} \dfrac{1-\operatorname{ctg}(\pi x)}{\operatorname{tg}(\pi x)-1} = \lim\limits_{x\to 1/4} \dfrac{\frac{\operatorname{tg}(\pi x)-1}{\operatorname{tg}(\pi x)}}{\operatorname{tg}(\pi x)-1} = \lim\limits_{x\to 1/4} \dfrac{1}{\operatorname{tg}(\pi x)} = 1$

8. $\lim\limits_{x\to 0} \dfrac{10^x-1}{2^x-1} = \lim\limits_{x\to 0} \dfrac{x\ln 10}{x\ln 2} = \lim\limits_{x\to 0} \dfrac{\ln 2+\ln 5}{\ln 2} = 1+\dfrac{\ln 5}{\ln 2}$

9. $\lim\limits_{x\to \frac{\pi}{2}} (\sin x)^{\operatorname{tg}^2 x} = \lim\limits_{x\to\frac{\pi}{2}} e^{\operatorname{tg}^2 x \ln\sin x} = \lim\limits_{x\to\frac{\pi}{2}} e^{\operatorname{tg}^2 x \ln(1+(\sin x-1))} =$

$= \lim\limits_{x\to\frac{\pi}{2}} e^{\operatorname{tg}^2 x(\sin x-1)} = \left| t = x-\frac{\pi}{2},\ t\to 0 \right| = \lim\limits_{t\to 0} e^{\operatorname{ctg}^2 t(\cos t-1)} = \lim\limits_{t\to 0} e^{-\frac{t^2}{2}\cdot\frac{1}{t^2}} = e^{-\frac{1}{2}}$

10. $\lim_{x \to -\infty} \operatorname{arccsin}\left(\sqrt{x^2+x} - x\right) = \lim_{x \to -\infty} \operatorname{arccsin}\left(\frac{x}{\sqrt{x^2+x} + x}\right) =$

$= \lim_{x \to -\infty} \left(\frac{1}{\sqrt{1+\frac{1}{x}} + 1}\right) = \frac{1}{2}$